Derivative

Definition

f is said to be differentiable at a point $a \in \mathbb{R}$, $\Leftrightarrow$ f is defined on open interval I containing a and:

$f'(a) := \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$ exists, $f'(a)$ is call derivative of f at a

remark: can also written as: $\lim_{x\to a} \frac{f(x)-f(a)}{x-a}$

differentiability is a local property, like continuity

Theorem

f is differentiable at a $\Leftrightarrow$ $\exists\, g: I \to \mathbb{R}$:

(1) g is continuous at a.

(2) $f(x) = g(x)(x-a) + f(a) \quad \forall\, x \in I$.

In this case: $f'(a) = g(a)$

proof: ① $g(x) = \begin{cases} \frac{f(x)-f(a)}{x-a}, & x \neq a. \\ f'(a), & x = a. \end{cases}$

Then $f(x) = (x-a)g(x) + f(a). \quad \forall\, x \in I$

$\lim_{x\to a} g(x) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a} = f'(a) = g(a) \Rightarrow$ g is cont. at a.

② $f(x) - f(a) = g(x)(x-a)$.

$f'(a) = \lim_{x\to a} \frac{f(x)-f(a)}{x-a} = \lim_{x\to a} g(x) = g(a)$. $\Rightarrow$ f is differentiable at a, $f'(a) = g(a)$

Theorem: if f is differentiable at a, then f is continuous at a.

proof: f is diff. at a $\Rightarrow$ $f(x) - f(a) = g(x)(x-a)$, for some g cont. at a.

$\lim_{x\to a}[f(x) - f(a)] = \lim_{x\to a} g(x)(x-a) = \lim_{x\to a} g(a)(a-a) = 0$.

$\therefore \lim_{x\to a} f(x) = f(a)$. f is cont. at a.

Endpoint Derivative

$f'_+(a) = \lim_{x\to a^+} \frac{f(x)-f(a)}{x-a}$ $\quad f'_-(\beta) = \lim_{x\to \beta^-} \frac{f(x)-f(\beta)}{x-\beta}$ "one-sided limit"

Definition:

A function f is differentiable on I $\Leftrightarrow$ $f'_+(a) = \lim_{\substack{x\to a \\ x\in I}} \frac{f(x)-f(a)}{x-a}$ exists and finite. $\forall\, a \in I$.

A function f is continuously differentiable on I if f'_I exists & cont. on I.

Standard Function Space

- $C^0(I)$: $\{ f : I \to \mathbb{R} :$ f is cont. on $I \}$
- $C^1(I)$: $\{ f : I \to \mathbb{R} :$ f' exist & cont. on $I \}$
- $\vdots$
- $C^{(n)}(I)$: $\{ f : I \to \mathbb{R} :$ $f^{(n)}$ exist & cont. on $I \}$
- $C^{(\infty)}(I)$: $\{ f : I \to \mathbb{R} :$ $f^{(n)}$ exist & cont. on I for all $n \in \mathbb{N} \}$

$\Rightarrow$ $C^0(I) \supseteq C^1(I) \supseteq \cdots \supseteq C^{(n)}(I) \supseteq C^{(n+1)}(I)$

$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}), & x \neq 0 \\ 0, & x = 0 \end{cases}$

(1) f is diff. on $\mathbb{R}$.

(2) f is not continuously diff at 0. $\Rightarrow f \notin C(\mathbb{R})$.

proof: (1) $\forall\, x \neq 0$. $f'(x) = 2x\sin(\frac{1}{x}) + x^2\cos(\frac{1}{x})\cdot\frac{-1}{x^2}$

$= 2x\sin(\frac{1}{x}) - \cos(\frac{1}{x})$

f is diff. at any $x \neq 0$

$f'(0) := \lim_{x\to 0} \frac{f(x)-f(0)}{x-0}$

$= \lim_{x\to 0} \frac{f(x)}{x} = \lim_{x\to 0} x\sin(\frac{1}{x}) = 0$.

$\therefore$ $f(x)$ is diff at all $x \in \mathbb{R}$ $\Rightarrow$ $f(x)$ is diff. on $\mathbb{R}$.

(2). $\lim_{x\to 0} f'(x) = f'(0)$?

$\lim_{x\to 0} (2x\sin(\frac{1}{x}) - \cos(\frac{1}{x}))$ $=$ $\lim_{x\to 0}(-\cos(\frac{1}{x}))$ DNE.

$\therefore$ f is not cont. diff. at 0

e.g. $f(x) = |x|$.

(1) f is diff. on $[0,1]$.

(2) f is diff. on $[-1,0]$

(3) f is not diff. on $[-1,1]$ $\quad f'_+(0) = \lim_{x\to 0^+} \frac{f(x)-0}{x-0} = 1$

(2). on $(0,1]$ $f(x) = x$, $f'(x) = 1$ exists $\Rightarrow$ f is diff. on $[0,1]$

(1) on $[-1,0)$ $f(x) = -x$ $f'(x) = -1$ exists $f'_-(0) = \lim_{x\to 0^-} \frac{f(x)-0}{x-0} = -1$ $\Rightarrow$ f is diff. on $[-1,0]$

(3). at $x=0$. $\lim_{x\to 0^+} \frac{f(x)-0}{x-0} = 1$.

$\lim_{x\to 0^-} \frac{f(x)-0}{x-0} = -1$. $\Rightarrow$ $\lim_{x\to 0} \frac{f(x)-0}{x-0} = f'(0)$ DNE., f is not diff. at $x=0$

Hence f is not diff on $[-1,1]$

Thm (Arithmetic): $f, g : I \to \mathbb{R}$. $a \in I$.

f, g are diff at a $\Rightarrow$ so is $f \pm g$, fg, αf ($\alpha \in \mathbb{R}$) at a

$\Rightarrow$ if $g(a) \neq 0$, so is $\frac{f}{g}$.

proof: $(fg)'(a) = \lim_{x\to a} \frac{f(x)g(x) - f(a)g(a)}{x-a}$.

(product).

$= \lim_{x\to a} \frac{f(x)g(x) - f(x)g(a) + f(x)g(a) - f(a)g(a)}{x-a}$

$= \lim_{x\to a} \frac{g(x)(f(x)-f(a))}{x-a} + \lim_{x\to a} \frac{f(a)(g(x)-g(a))}{x-a}$.

$= g(a)f'(a) + f(a)g'(a)$

Quotient: $(\frac{1}{f})'(a) = \lim_{x\to a} \frac{\frac{1}{f(x)} - \frac{1}{f(a)}}{x-a}$

$= \lim_{x\to a} \frac{g(a)-g(x)}{(x-a)f(x)g(a)}$

$= -\frac{1}{f(a)} \lim_{x\to a} \frac{g(x)-g(a)}{x-a}$

$= -\frac{g'(a)}{g(a)^2}$

$\therefore (\frac{1}{f})'(a) = (1\cdot\frac{1}{f})'(a)$

Chain Rule

Suppose f is diff. at a and g is diff. at $f(a)$, then $g\circ f$ is diff. at a

$(g\circ f)'(a) = g'(f(a))\cdot f'(a)$

proof: Since f is diff at a. $\exists\, F$: $f(x) = F(x)(x-a) + f(a)$.

g is diff. at $f(a)$ $\exists\, G$: $g(y) = G(y)(y - f(a)) + g(f(a))$

$g(f(x)) - g(f(a)) = G(f(x))(f(x) - f(a))$

Since f is diff. at a. f is cont at a.

$(g\circ f)'(a) = \lim_{x\to a} \frac{g(f(x)) - g(f(a))}{x-a} = \lim_{x\to a} G(f(x)) \frac{f(x)-f(a)}{x-a} = G(f(a))F(a) = g'(f(a))f'(a)$

Rolle's Theorem

Let $a < b$. if f is Cont. on $[a,b]$, diff on (a,b). $f(a) = f(b)$.

then $\exists\, c \in (a,b)$ $f'(c) = 0$.

By Extreme Value Theorem

f has a maximum M and min m in $[a,b]$

if $m = M$: f is constant. $f(x) = 0$. $\forall\, x \in (a,b)$

if $m < M$: since $f(a) = f(b)$ $\exists\, c \in (a,b)$ $f(c) = m$ or M.

Suppose $f(c) = m$. Then $f(c+h) \geq f(c)$ $\forall$ sufficient small h.

let $\{x_n\}$ be Seq. such that $x_n \to c$ and $x_n > c$. $\Rightarrow$ $\lim_{n\to\infty} \frac{f(x_n)-f(c)}{x_n - c} \geq 0$.

$\Vert$

$f'(c)$.

let $\{y_n\}$ be Seq. such that $y_n \to c$. and $y_n < c$

$\Rightarrow$ $\lim_{n\to\infty} \frac{f(y_n)-f(c)}{y_n - c} \leq 0$.

$\Vert$

$f'(c)$.

$\Rightarrow$ $f'(c) \geq 0$ and $f'(c) \leq 0$ $\Rightarrow$ $f'(c) = 0$

Mean Value Theorem

if f is cont. on $[a,b]$ and diff on (a,b), then there is $c \in (a,b)$ such that

$f'(c) = \frac{f(b)-f(a)}{b-a}$, $\quad f'(c)(b-a) = f(b) - f(a)$

proof: set Secant line

$l(x) = \frac{f(b)-f(a)}{b-a}(x-a) + f(a)$

Let $h(x) = f(x) - l(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a)$.

$h(a) = h(b) = 0$ $\Rightarrow$ $\exists\, c \in (a,b)$. $h'(c) = 0$.

$\Rightarrow$ $h'(c) = f'(c) - \frac{f(b)-f(a)}{b-a} = 0$.

$\Rightarrow$ $f'(c) = \frac{f(b)-f(a)}{b-a}$

Monotone functions

Monotone functions

Let $E \subseteq \mathbb{R}$ be nonempty and $f : E \to \mathbb{R}$.

(i) The function f is *increasing* on E if $x_1 < x_2$ implies $f(x_1) \leq f(x_2)$; it is *strictly increasing* if the inequality is strict.

(ii) It is *decreasing* on E if $x_1 < x_2$ implies $f(x_1) \geq f(x_2)$; it is *strictly decreasing* if the inequality is strict.

(iii) It is *monotone* if it is either increasing or decreasing.

For example, x^2 is monotone on $[-1,0]$ and on $[0,1]$, but not on $[-1,1]$.

Derivative test for monotonicity

Suppose f is continuous on $[a,b]$ and differentiable on (a,b).

(i) If $f'(x) \geq 0$ on (a,b), then f is increasing on $[a,b]$; if $f'(x) > 0$, then f is strictly increasing.

(ii) If $f'(x) \leq 0$ on (a,b), then f is decreasing; if $f'(x) < 0$, then f is strictly decreasing.

(iii) If $f'(x) = 0$ on (a,b), then f is constant on $[a,b]$.

proof if $x_1 < x_2$ MVT: $\exists\, \theta \in (x_1, x_2)$ $f'(\theta) = \frac{f(x_2)-f(x_1)}{x_2 - x_1}$

$f(x_2) - f(x_1) = f'(\theta)(x_2 - x_1)$

Since $x_2 - x_1 > 0$, $f(x_2) - f(x_1) > 0$ or < 0 only depends on $f'(x)$

$\Rightarrow$ $\begin{cases} f'(x) > 0 \Rightarrow f(x_2) > f(x_1) \Rightarrow \text{increasing} \\ \cdots \end{cases}$

Thm 4.18

Suppose f is increasing on $[a,b]$

① if $c \in [a,b)$ then $f(c+)$ exists and $f(c) \leq f(c+) = \lim_{x\to c^+} f(x)$.

② if $c \in (a,b]$ then $f(c-)$ exists and $f(c-) \leq f(c)$

③ if $c_1 < c < c_2$ then

$f(c_1) \leq f(c-) \leq f(c) \leq f(c+) \leq f(c_2)$

proof (2): $E = f([a,c))$ $\quad s = \sup E = \sup_{x\in[a,c)} f(x)$.

$f(x) \leq f(c)$ for $x < c$ $\Rightarrow$ $s \leq f(c)$.

$\forall\, \varepsilon > 0$. choose $x_\varepsilon < c$. $s - \varepsilon < f(x_\varepsilon) \leq s$

$\Rightarrow$ For $x_\varepsilon < x < c$. $s - \varepsilon < f(x_\varepsilon) \leq f(x) \leq s$

$|f(x) - s| < \varepsilon$ $\Rightarrow$ $f(c-) = s \leq f(c)$

Theorem 4.19. if f is monotone on an interval I.

then the set $\{x \in I : f$ is not cont. at $c\}$ is at most Countable.

4.21 Bernoulli theorem

Let $\alpha > 0$ and $x \in [-1, +\infty)$

(i) if $0 < \alpha < 1$ then $(1+x)^\alpha \leq 1 + \alpha x$.

(ii) if $\alpha \geq 1$ then $(1+x)^\alpha \geq 1 + \alpha x$

proof: using mean value theorem on t^α between 1 and $x+1$

$\exists\, c$: $\alpha c^{\alpha-1} = \frac{(x+1)^\alpha - 1^\alpha}{x} = (x+1)^\alpha - 1$.

$\alpha x c^{\alpha-1} = (x+1)^\alpha - 1$

if $\alpha \in (0,1]$. if $x \geq 0$, then $c \geq 1$. $c^{\alpha-1} \leq 1 \Rightarrow \alpha x c^{\alpha-1} \leq \alpha x$

if $-1 < x < 0$ then $0 < c < 1$. $c^{\alpha-1} \geq 1 \Rightarrow \alpha x c^{\alpha-1} \leq \alpha x$

$(x+1)^\alpha - 1 \leq \alpha x \Rightarrow (1+x)^\alpha \leq 1 + \alpha x$

if $\alpha \geq 1$ $\quad (1+x)^\alpha \geq 1 + \alpha x$

Thm 4.23. IVT For Derivatives

Suppose f is diff. on $[a,b]$ with one-sided derivatives at endpoints

and $f'(a) \neq f'(b)$. if y_0 (lies strictly between $f'(a)$ & $f'(b)$) then there

is an $x_0 \in (a,b)$ such that $f'(x_0) = y_0$

Assume $f'(a) < y_0 < f'(b)$ set $F(x) = f(x) - y_0 x$. EVT.: F attains minimum at some $x_0 \in [a,b]$

$F(a+h) = f'(a) - y_0 < 0$ for all sufficiently small $h > 0$ we have $F(a+h) < F(b)$. So a is not minimum point.

Similarly, b is not minimum point. Hence $x_0 \in (a,b)$ Fermat theorem $F'(x_0) = f'(x_0) - y_0 = 0$. $\Rightarrow$ $f'(x_0) = y_0$. $x_0 \in (a,b)$

Taylor's Theorem & L'Hôpital's Rule

1. Generalized MVT

if f, g are cont. on $[x, x_0]$ and diff. on (x, x_0). then there is a c between x_0, x such that

$(f(x) - f(x_0))\, g'(c) = (g(x) - g(x_0))\, f'(c)$.

take $g(t) = t$ gives the usual MVT. $f'(c) = \frac{f(x)-f(x_0)}{x - x_0}$.

Also. if f is diff. at x_0 where it has a local max. or min. then $f'(x_0) = 0$.

proof ① $\exists\, c$: $\begin{cases} f'(c) = \frac{f(x)-f(x_0)}{x-x_0} \\ g'(c) = \frac{g(x)-g(x_0)}{x-x_0} \end{cases}$ $\Rightarrow$ $\frac{f'(c)}{g'(c)} = \frac{f(x)-f(x_0)}{g(x)-g(x_0)}$ $\Rightarrow$ $f'(c)[g(x)-g(x_0)] = g'(c)[f(x)-f(x_0)]$.

② if $f(x_0) > 0$ and x_0 is a local maximum

then $\exists$ sufficiently small $h > 0$: $\frac{f(x_0+h)-f(x_0)}{h} > 0$, contradicting maximality

similarly to minimality

One-sided limits of monotone functions.

if f is increasing on I & c is a interior point. then $f(c-)$ & $f(c+)$ exists

Moreover whenever $a_1 < c < c_2$ on I.

$f(c_1) \leq f(c-) \leq f(c) \leq f(c+) \leq f(c_2)$

proof: Let $E = \{f(x) : x \in I,\ x < c\}$. $\sup E = s$.

$\forall\, \varepsilon > 0$. choose $x_\varepsilon < c$ such that $s - \varepsilon < f(x_\varepsilon) \leq s$, since f is increasing.

$\forall\, x \in (x_\varepsilon, c)$. $s - \varepsilon < f(x_\varepsilon) \leq f(x) \leq s$ $\Rightarrow$ $|f(x) - s| < \varepsilon$ $\Rightarrow$ $\lim_{x\to c^-} f(x) = f(c-) = s$.

$\Rightarrow$ $\forall\, c_1 < c$. $f(c_1) \leq s = f(c-)$.

Similarly we can use infimum to prove $f(c+) \geq f(c)$

Taylor's Formula

$n \in \mathbb{N}$ $(\{0, 1, 2, \cdots\})$, $a < b$ $\quad f : (a,b) \to \mathbb{R}$ has $(n+1)$st derivative on (a,b). For every $x, x_0 \in (a,b)$ there is a c between x & x_0 such that:

$f(x) = \sum_{k=0}^{n} \frac{f^k(x_0)}{k!}(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$.

$P_n^{f,x_0}(x) := \sum_{k=0}^{n} \frac{f^k(x_0)}{k!}(x - x_0)^k$ is the Taylor polynomial of order n for f. centered at x_0.

$\Rightarrow$ $f(x) - P_n^{f,x_0}(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$.

$n = 0$: MVT. $\quad f(x) - f(x_0) = f'(c)(x - x_0)$

$n = 1$. $\quad f(x) - f(x_0) - f'(x_0)(x - x_0) = \frac{f''(c)}{2}(x - x_0)^2$.

$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(c)}{2}(x - x_0)^2$.

e.g. $f(x) = e^x$. $\quad x_0 = 0$.

$P_n^{f,x_0}(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$.

$= \sum_{k=0}^{n} \frac{1}{k!}(x)^k = \sum_{k=0}^{n} \frac{x^k}{k!}$

$\Rightarrow$ $e^x - \sum_{k=0}^{n} \frac{x^k}{k!} = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1} = \frac{e^c}{(n+1)!}x^{n+1}$ $\quad c \in (0, x)$.

if $x \in [-1, 1]$ $\Rightarrow$ $|e^x - \sum_{k=0}^{n} \frac{x^k}{k!}| \leq \frac{e}{(n+1)!} \leq \frac{3}{(n+1)!}$,

$n \to \infty$ $\Rightarrow$ $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

Proof of Theorem 4.24. Assume $x_0 < x$; the other case is analogous. For $t \in [x_0, x]$, define

$$F(t) = \frac{(x-t)^{n+1}}{(n+1)!}$$

and

$$G(t) = f(x) - f(t) - \sum_{k=1}^{n} \frac{f^{(k)}(t)}{k!}(x-t)^k.$$

Differentiating each term in the sum produces a telescoping cancellation:

$$G'(t) = -\frac{f^{(n+1)}(t)}{n!}(x-t)^n, \qquad F'(t) = -\frac{(x-t)^n}{n!}.$$

The generalized Mean Value Theorem gives $c \in (x_0, x)$ such that

$$(F(x) - F(x_0))G'(c) = (G(x) - G(x_0))F'(c).$$

Since $F(x) = G(x) = 0$, we obtain

$$G(x_0) = F(x_0)\frac{G'(c)}{F'(c)} = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1},$$

L'Hôpital's Rule

$f(x) \to 0$ & $g(x) \to 0$, or $|f(x)| \to \infty$ & $|g(x)| \to \infty$.

$\Rightarrow$ $\lim_{\substack{x\to a \\ x\neq 1}} \frac{f(x)}{g(x)} = \lim_{\substack{x\to a \\ x\in I}} \frac{f'(x)}{g'(x)}$

Inverse function Theorem

$f : I \to \mathbb{R}$. be one-to-one & cont. then:

(i) $J = f(I)$ is an interval.

(ii) f is Strictly monotone

(iii) $f^{-1} : J \to I$ is Continuous & monotone on J

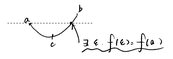
proof: (i) if $c < y < d$ are points on J. Choose $u, v \in I$ $\Rightarrow$ $f(u) = c$, $f(v) = d$.

intermediate value theorem: $\exists\, e \in (u, v)$, $f(e) = y$ $\Rightarrow$ J is a interval.

(ii) if f is not Strictly monotone: $\exists\, a < c < b$ which $f(c)$ does not lie between $f(a)$, $f(b)$.

$\Rightarrow$ IVT.: there will be a second preimage of either $f(a)$ or $f(b)$

Contradicted to injectivity.



(iii) if f is Strictly increasing then f^{-1} is Strictly increasing.

- **反函数连续：**假设 f 递增，则 f^{-1} 也递增。取 $y_0 = f(x_0)$，在 x_0 两侧选足够近的点 $x_- < x_0 < x_+$。当 y 充分靠近 y_0 时，单调性迫使 $f^{-1}(y)$ 落在 x_- 与 x_+ 之间，因此 $f^{-1}(y)$ 靠近 x_0。在区间端点，去掉区间外不存在的那一侧界即可。递减情形类似。

Theorem 4.33

$f : I \to \mathbb{R}$ is one-to-one & cont. if $a \in I$. $f(a) = b$., $f'(a)$ exists and is nonzero. then f^{-1} is diff. at b.

$(f^{-1})'(b) = \frac{1}{f'(a)}$.

proof: $(f^{-1})'(b) = \lim_{y\to b} \frac{f^{-1}(y) - f^{-1}(b)}{y - b} = \lim_{x\to a} \frac{x - a}{f(x) - f(a)} = \frac{1}{f'(a)}$.